

Semantic segmentation - attempts at three different combinations

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Abstract—This article aims to develop and compare different computer vision methods to achieve semantic segmentation of natural environment images. We tested the segmentation performance of these methods in natural environment images using the 2D portion of the WildScenes dataset. Specifically, this article reproduced the DeepLabV3 model and conducted experiments using ResNet-50 and MobileNetV3 as its backbone networks. Meanwhile, we also adopted the SegNet model and set its backbone network as ResNet-50. Through experiments, we compared the average intersection to union (mIoU) values of these three methods in different categories and found that the DeepLabV3-resnet-50 model performed better than the other two models overall. We analyzed and compared the results to propose the reasons for this phenomenon.

Keywords—Semantic segmentation, natural environment images, computer vision, DeepLabV3, ResNet-50, MobileNetV3, SegNet, WildScenes dataset, deep learning, image processing

I. INTRODUCTION

Semantic segmentation is an important task in computer vision, which involves classifying each pixel in an image into different semantic categories. This classification not only involves identifying objects in the image, but also involves precise classification and labeling of each pixel, thereby achieving a deep understanding of the image content. For example, in autonomous driving, semantic segmentation can help vehicles recognize roads, pedestrians, traffic signs, etc., thereby improving driving safety. Nowadays, autonomous driving is gradually making breakthroughs in urban environments, but how to accurately navigate in natural environments is still full of challenges. In natural environments, objects of different categories may have very similar appearance features, such as different types of trees or flowers. Objects in the natural environment often obstruct each other, such as leaves covering some animals, or stones covering some plants. The goal of this project is to develop and compare different computer vision methods for semantic segmentation of natural environment images. This project used the 2D portion of the WildScenes dataset to test the segmentation performance of different methods on images in natural environments. The

WildScenes dataset is a benchmark dataset specifically designed for large-scale 2D and 3D semantic segmentation tasks in natural environments. This dataset provides high-resolution 2D images and high-density 3D LiDAR point clouds, accompanied by precise six degrees of freedom (6-DoF) pose information (Kavisha et al. (2023)). The data was collected in two different natural environments in Australia (Venman and Karawatha) (Kavisha et al. (2023)). We replicated the DeepLabV3 model used in the papers supported by the dataset and replaced its backbone. Secondly, we used the SegNet method and observed the IoU values generated by various methods to determine the quality of the methods.

II. LITERATURE REVIEW

A. DeepLabv3

DeepLab is a series of deep learning models developed by the Google research team for semantic segmentation tasks. The goal of semantic segmentation is to assign each pixel in an image to a category, and DeepLab improves the performance of this task by introducing and improving some key techniques. The DeepLab series of models aims to address multiple challenges in semantic segmentation tasks, including processing multi-scale information and precise segmentation of high-resolution images. The design and development of DeepLab has gone through several major versions, each of which has made improvements based on the previous version. In this group project task, we select DeepLabv3 as implementation method.

DeepLab system introduces Atrous Convolution, a technique that expands the receptive field by introducing intervals in the convolution operation without increasing computational complexity or reducing spatial resolution. This can better capture contextual information in the image. DeepLabv3 system improves the use of dilated convolutions by adopting Dilated Separable Convolution, further enhancing computational efficiency and accuracy. Removing CRFs and achieving better refinement and multi-scale feature fusion through improved dilated convolution.

DeepLabv3 system significantly improves over DeepLab versions and attains comparable performance with other state-

of-art models on the semantic image segmentation benchmark. The proposed model “DeepLabv3” employs atrous convolution with upsampled filters to extract dense feature maps and to capture long range context. Compared to previous DeepLab versions, this model has made significant improvements and achieved performance comparable to other state-of-the-art models on semantic image segmentation benchmarks (Chen et al. (2017)).

B. ResNet

ResNet is a deep convolutional neural network architecture, abbreviated as “Residual Networks”. Its design goal is to solve the degradation problem encountered by deep neural networks during training (i.e., the problem of increasing training error as the network depth increases).

The core innovation of ResNet is the introduction of Residual Blocks. Each residual block contains one or more convolutional layers and a shortcut connection, which directly adds the input to the output. This design enables the network to learn residuals (i.e. the difference between the actual output and the target output), making the optimization process more efficient. ResNet-50 is a specific version of the ResNet series, where “50” indicates that there are 50 layers in the network. Compared to shallow networks, ResNet-50 can capture more complex features, resulting in better performance in various visual tasks. ResNet-50 is a classic and practical deep learning model widely used in the field of image processing, and its design concept of residual connections has had a profound impact on many subsequent neural network architectures.

ResNet provides a residual learning framework to simplify deeper network training than previously used networks. It does not learn functions that are not referenced, but explicitly rephrases the layers as learning residual functions input by the reference layer. It provides comprehensive empirical evidence that these residual networks are easier to optimize and can achieve accuracy from significantly increased depths. When deeper networks can begin to converge, a degradation problem is exposed: as the depth of the network increases, accuracy saturates (although this is not surprising), and then rapidly degrades. The decrease in training accuracy indicates that not all systems are equally easy to optimize. It solves the degradation problem by introducing a deep residual learning framework (He et al. (2016)).

C. MobileNet

MobileNet is a series of lightweight deep learning models developed by Google, specifically optimized for computer vision tasks on mobile and embedded devices. These models are highly efficient in terms of processing speed and computational resource consumption, while still maintaining good accuracy. The MobileNet series models are widely used in tasks such as image classification, object detection, and semantic segmentation.

MobileNet Proposed the concept of depth wise separable convolution. This is a technique that decomposes standard convolution operations into deep convolution and pointwise convolution. This method significantly reduces the computational complexity and number of parameters while maintaining the expressive power of the model. MobileNetV3 is the latest version of the MobileNet series, inheriting the

advantages of the previous two versions and further optimizing efficiency and performance. It introduces new design concepts and technologies to meet modern computing needs. MobileNetV3 is further optimized on the basis of MobileNetV1 by using AutoML technology to design more efficient network structures and introducing hardware friendly activation functions and modules (such as SE module and H-swish activation function).

MobileNetV3 combines hardware and software network architecture search (NAS), combined with the NetAdapt algorithm, and then improves through novel architecture advancements to match mobile CPUs. There are two models of MobileNetV3: MobileNetV3 Large and MobileNetV3 Small. These models are designed for high resource and low resource use cases respectively. In practical applications, if a certain balance needs to be struck between efficiency and accuracy, such as real-time video processing, image segmentation tasks on mobile applications and embedded devices, MobileNetV3 is a good choice (Howard et al. (2019)).

D. SegNet

SegNet is a deep convolutional neural network (CNN) architecture used for semantic segmentation tasks. It is mainly used to divide images into regions with specific semantic categories. The design of SegNet considers how to improve computational efficiency while ensuring segmentation accuracy. The architecture of SegNet consists of encoder, decoder, and classifier. The main features are pooling indices transfer and pixel by pixel prediction.

The core trainable segmentation engine of SegNet consists of an encoder network, a corresponding decoder network followed by a pixel-wise classification layer. The novelty of SegNet lies in the way the decoder upsamples its lower resolution input feature maps. The main motivation of SegNet is scene understanding applications. Therefore, its memory consumption and computation time are relatively low. Compared to other competitive architectures, it has much fewer trainable parameters and can be trained end-to-end using stochastic gradient descent (Badrinarayanan et al. (2017)).

E. Our project select three methods to perform semantic segmentation tasks

- (1)DeepLabV3_resnet-50(DeepLabV3 based on ResNet-50);
- (2)DeepLabV3_MobileNetV3(DeepLabV3 based on MobileNetV3);
- (3)Seg_ResNet-50(a simple image segmentation model based on ResNet-50).

DeepLabv3_resnet-50 is based on ResNet-50 network. Introduced dilated convolution to expand the receptive field of the convolution kernel without increasing computational complexity, thus better capturing contextual information. We used Spatial Pyramid Pooling (ASPP) to obtain multi-scale contextual information, which improved the segmentation performance of the model for different scales and backgrounds.

DeepLabV3-MobileNetV3 is a model that combines the efficient semantic segmentation strategy of DeepLabV3 with the lightweight feature extraction network of MobileNetV3. This

combination aims to achieve efficient and accurate semantic segmentation, especially suitable for running on resource constrained devices. DeepLabV3nMobileNetV3 combines the powerful segmentation capabilities of DeepLabV3 with the efficient feature extraction of MobileNetV3.

Seg_ResNet-50 is based on ResNet-50 network. Mainly used to generate feature maps, which are then used for image segmentation. Add some additional layers based on ResNet-50, such as upsampling layers and decoders, for image segmentation. The processing method is relatively simple and suitable for basic image segmentation tasks.

III. METHODS

A. DeepLabV3-ResNet50

We replicated the DeepLabV3 model mentioned in the dataset paper and the encoder backbone was ResNet50. DeepLabV3 is an advanced image semantic segmentation model. DeepLabV3 mainly consists of Atrous Convolution, Atrous Spatial Pyramid Pooling (ASPP), Batch Normalization, and Depth wise Separable Convolution .Atrous Convolution expands the receptive field of the convolution kernel by inserting "holes" into it, without increasing computational complexity and this enables the model to capture a larger range of contextual information while maintaining high-resolution feature maps; ASPP uses multi-scale dilated convolution and global average pooling to fuse features of different scales, enhancing the model's ability to detect multi-scale targets; Batch Normalization and Depth wise Separable Convolution further improve computational efficiency and reduce the computational burden on the model (Chen et al.(2018)).

B. DeepLabV3-MobileNetV3

Afterwards, we replaced the backbone in the DeepLabV3 model with MobileNetV3 and conducted training. MobileNetV3 is a lightweight convolutional neural network that is suitable as a backbone for various computer vision tasks. It has efficient performance on mobile and embedded devices. MobileNetV3 achieves higher computational efficiency and lower latency through optimized depth wise separable convolutions and new activation functions such as h-swift(Howard et al.(2019)). Secondly, the accuracy and performance of the model are balanced. MobileNetV3 optimizes its network structure through Neural Architecture Search (NAS). NAS automatically searches for the optimal network structure to ensure efficient performance across devices with limited resources. (Howard et al.). In situations where resources are limited (such as drones, autonomous vehicles, etc.) MobileNetV3 is a good choice, and by combining some methods to improve semantic segmentation accuracy, such as the DeepLabV3 we use, MobileNetV3 can further enhance its performance in semantic segmentation tasks.

C. SegNet-ResNet50

SegNet is a deep learning neural network model used for image segmentation, particularly suitable for semantic segmentation tasks. The core segmentation engine of this model consists of an encoder network, a decoder network, and a pixel level classification layer. The core segmentation engine of this model is composed of an encoder network, a matching decoder network, and a pixel-level classification layer (Badrinarayanan, V., Kendall, A., & Cipolla, R. (2017)). The innovation of

SegNet lies in the way the decoder upsamples its lower resolution input feature maps. Concretely, the decoder uses the pooling index calculated in the maximum pooling step of the corresponding encoder to perform non-linear upsampling. This way, there is no need to learn upsampling. The upsampled map is sparse, and then convolved with trainable filters to generate dense feature maps (Badrinarayanan, V., Kendall, A., & Cipolla, R. (2017)). We have chosen ResNet50 as the backbone of this model, since ResNet50 solves the problem of gradient vanishing in deep networks through residual connections, allowing the network to go deeper and extract richer features (He et al. (2016)).

IV. EXPERIMENTAL RESULTS

In our project task, we select AutoDL, a low-cost GPU computing platform. AutoDL is a cloud-based machine learning service platform that helps users quickly build and deploy machine learning models. We rented a GPU server on AutoDL for model training and computation.

The experimental setup we used is as follows.

A. Dataset

- Data sources: The dataset to be used in the group project is called WildScenes (see links and references at the end of this document). This is a recently released multimodal dataset consisting of five sequences of 2D images recorded with a normal video camera during traversals through two forests: Venman National Park and Karawatha Forest Park, Brisbane, Australia.

- Data description: The dataset has 9,306 images of size 2,016 x 1,512 pixels. Each and every one of the images has been manually annotated.

- Data preprocessing: Data normalization, data augmentation and transformation, including resizing, random horizontal flipping, random rotation, and color adjustment, etc;

- Label preprocessing: Including type conversion, resizing, and label mapping.

- Training/validation/testing division: Randomly divide into 70%, 5%, and 25% respectively.

B. Experimental environment

- Hardware configuration: AutoDL cloud computing GPU Processor

- Software environment: Windows operating system, Python programming language, Python IDE (PyCharm), machine learning libraries (PyTorch), Deep learning models (DeepLab, ResNet, MobileNet), etc.

C. Experimental steps

- Environment settings and logging: Set the log file path and configure the logger to record important information during the training process.

- Define category metadata: Define the names and color palettes of various categories in the dataset for easy analysis and visualization in the future.

- CUDA memory configuration: Set CUDA memory to be expandable to improve memory utilization efficiency.

- Label Mapping and Conversion: Define a label mapping function to convert specific category labels into target labels or ignore them; Data augmentation and image/label conversion, including resizing, random transformation, and standardization.

- Load the dataset: Instantiate training and validation datasets, specify paths and transformation methods for images and labels.

- Check the balance of the dataset: Count and record the sample size of each category to understand the balance of the dataset.

- Configure data loader: Create training and validation data loaders, set batch size and number of parallel processing worker threads.

- Model settings: Load various pre trained models and adjust the final classification layer to fit the number of categories in the dataset; Use the cross entropy loss function (ignoring pixels with index 255); Choose the Adam optimizer and set the learning rate to 0.001.

- Define loss function and optimizer: Use CrossEntropyLoss as the loss function and set ignore_index to ignore specific labels; Use Adam optimizer to optimize model parameters.

- Training and Verification Cycle: Conduct multiple epochs of training and validation.

- Model saving: After each epoch, save the model's state dictionary for subsequent loading and use.

- Visualization and Analysis: Draw the change curves of loss, accuracy, and IoU during the training and validation process, and save them as image files.

- Save results to CSV file: Calculate the final validation results, including the IoU for each category, and save them to a CSV file for further analysis.

- Draw plots and visualize.

Through experiments, we obtained the following results.

Based on DeepLabV3_resnet-50, we obtained the following segmentation results.

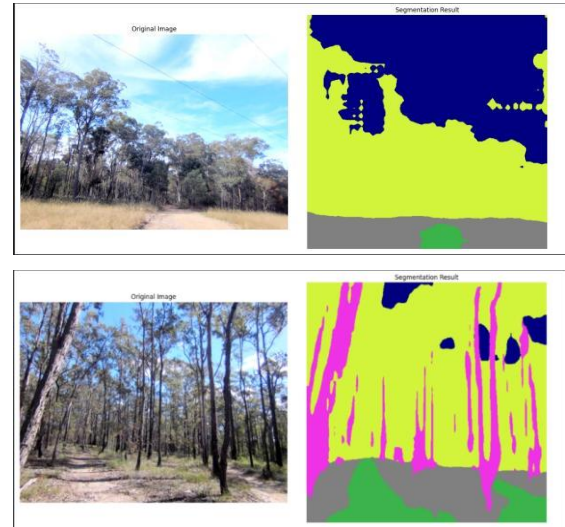


FIGURE 1. EXAMPLE SEGMENTATION RESULTS IN METHOD DEEPLABV3_RESNET-50

We get mIoU of 21.81.

Class	mIoU	dirt	mud	water
IoU	21.81	60.88	0.27	1.5
Class	gravel	other-terrain	tree-trunk	tree-trunk
IoU	6.46	1.56	41.78	81.43
Class	bush	fence	structure	rock
IoU	7.98	1.3	4.36	0.4
Class	log	other-object	sky	grass
IoU	3.26	1.67	60.39	53.86

TABLE 1. TABLE OF MIOU AND THE OTHER CLASS IOU IN METHOD DEEPLABV3_RESNET-50

The metrics curves of Loss, Accuracy and IoU are shown in the following figure.



FIGURE 2. FIGURE OF LOSS, ACCURACY AND IOU IN METHOD DEEPLABV3_RESNET-50

Based on DeepLabV3_MobileNetV3, we obtained the following segmentation results.

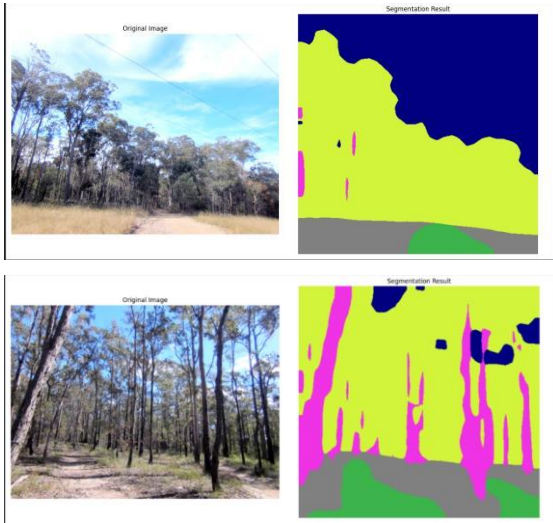


FIGURE 3. EXAMPLE SEGMENTATION RESULTS IN METHOD DEEPLABV3_MOBILENETV3

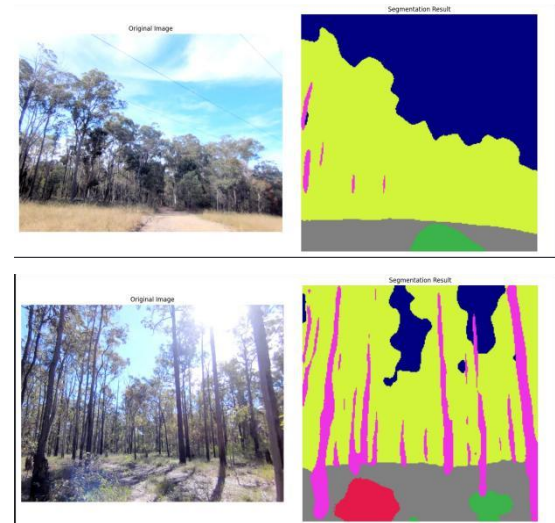


FIGURE 5. EXAMPLE SEGMENTATION RESULTS IN METHOD SEG_RESNET-50

We get mIoU of 20.84.

Class	mIoU	dirt	mud	water
IoU	20.84	55.52	21.50	1.26
Class	gravel	other-terrain	tree-trunk	tree-trunk
IoU	4.39	0.86	29.72	79.74
Class	bush	fence	structure	rock
IoU	6.3	0.83	3.11	0.6
Class	log	other-object	sky	grass
IoU	3.35	1.54	52.15	51.70

TABLE 2. TABLE OF mIoU AND THE OTHER CLASS IoU IN METHOD DEEPLABV3_MOBILENETV3

The metrics curves of Loss, Accuracy and IoU are shown in the following figure.

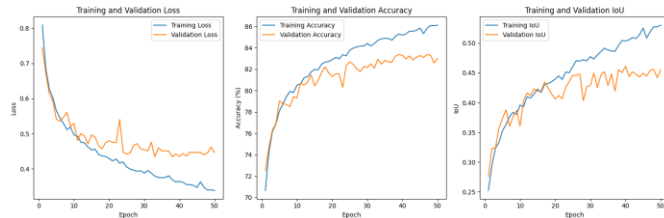


FIGURE 4. FIGURE OF LOSS, ACCURACY AND IOU IN METHOD DEEPLABV3_MOBILENETV3

Based on Seg_ResNet-50, we obtained the following segmentation results.

We get mIoU of 20.28.

Class	mIoU	dirt	mud	water
IoU	20.28	57.80	0	1.47
Class	gravel	other-terrain	tree-trunk	tree-trunk
IoU	3.76	1.06	35.13	82.9
Class	bush	fence	structure	rock
IoU	5.88	0	3.09	0
Class	log	other-object	sky	grass
IoU	0	0	57.6	55.44

TABLE 3. TABLE OF mIoU AND THE OTHER CLASS IoU IN METHOD SEG_RESNET-50

The metrics curves of Loss, Accuracy and IoU are shown in the following figure.

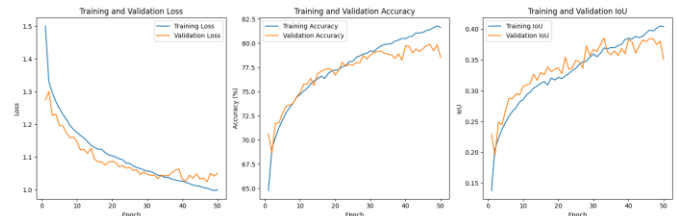


FIGURE 6. FIGURE OF LOSS, ACCURACY AND IOU IN METHOD SEG_RESNET-50

The data obtained indicates that, based on DeepLabV3_resnet-50, we can get the biggest mIoU. This Model demonstrates g

ood learning ability, but fluctuations in validation metrics suggest a possible slight overfitting; Based on DeepLabV3_resnet-50, We achieve the highest accuracy and all indicators are relatively stable; Based on Seg_ResNet-50, We find that the model performs reasonably in all aspects, but has significant fluctuations and weak generalization ability.

In summary, Model DeepLabV3_resnet-50 appears to perform the best, with the highest accuracy and IoU, and relatively stable validation metrics, indicating good generalization ability.

V. DISCUSSION

A. DeepLabV3-ResNet50

The DeepLabV3-ResNet50 combination is a widely used deep learning architecture for semantic segmentation tasks. It combines the advanced feature extraction technology of DeepLabV3 with the powerful residual network structure of ResNet50. DeepLabV3 uses dilated convolutions to expand the receptive field and capture richer contextual information without increasing computational complexity (Chen et al. (2017)). The residual structure of ResNet50 enables the network to be deeper, thereby improving its feature extraction capability (He et al. (2016)). However, post training analysis shows that the model still has the following shortcomings. Based on relevant research, we have found that although dilated convolution reduces computational complexity to some extent, the DeepLabV3-ResNet50 combination still requires high computational resources overall, especially when processing high-resolution images (Chen et al. (2017)) (He et al. (2016)). Due to the use of network depth and dilated convolutions, the model consumes a significant amount of video memory during training and inference (Chen et al. (2017)).

B. DeepLabV3-MobileNetV3

We chose the MobileNetV3 model based on its efficiency and lightweight advantages, as well as its ability to achieve higher accuracy when resources are limited. However, in fact, through experimental research results, we found that the model cannot effectively segment images with higher accuracy and get better mIoU value (Some classes have high IoU values, while others have low or even zero IoU values). The reason for this result is firstly Due to the small parameter and computational complexity of MobileNetV3, it may not be able to provide performance comparable to more complex networks when dealing with complex visual tasks such as high-resolution image classification, object detection, and semantic segmentation (Sandler et al. (2018)). Secondly, when we trained the model according to task requirements, we only used a portion of the images (K01, V01), so the generalization ability may be insufficient.

C. SegNet-ResNet50

We chose the SegNet model since SegNet significantly reduces memory consumption by storing only max pooling indices during the pooling process. In addition, this model using pooling index for upsampling can effectively restore the spatial information of the image, making the segmentation results more refined (Badrinarayanan, V., Kendall, A., & Cipolla, R. (2017)). During the training process, we found that the model was able to get good IoU values but did not achieve very high mIoU values during the testing process. We assume that this problem may stem from the selection of the dataset. Specifically, the limited subset of the dataset used may lead to a decrease in the model's ge-

neralization ability. In addition, SegNet uses a deep convolutional neural network (CNN) architecture, which typically contains many convolutional layers and parameters. During the training process, these layers and parameters require a significant amount of computation and optimization, resulting in longer training time and more computing resources.

VI. CONCLUSION

A. DeepLabV3-ResNet50

What worked and did not work: The training and validation losses have steadily decreased, but the validation loss fluctuates greatly, which may indicate that the model has some overfitting. The training accuracy has increased to about 82%, while the validation accuracy has reached around 78%, indicating that the model has good learning ability but still has room for improvement. Both the training and validation of IoU have steadily increased, but fluctuations in the validation of IoU indicate instability in the model when dealing with different samples.

Future work: Adopt stronger regularization methods, such as a higher L2 regularization coefficients or increasing Dropout rates, to alleviate overfitting. Further expand the types of data augmentation, such as scaling, padding, etc., to improve the robustness of the model.

B. DeepLabV3-MobileNetV3

What worked and did not work: The decreasing trend of training and validation losses is good, and the validation loss is close to the training loss, indicating that the model has good generalization ability. The training accuracy reached about 86%, and the validation accuracy reached around 82%, which is the highest among the three models. This indicates that the model has achieved a good balance between accuracy and efficiency. Both the training and validation of IoU are high, and the validation of IoU shows small fluctuations, demonstrating the stable performance of the model on diverse datasets.

Future work: Trying different optimizers or adjusting learning rate strategies, such as learning rate schedulers, may further improve performance. The lightweight feature of MobileNetV3 makes it suitable for devices with limited resources, but it may lead to insufficient feature expression ability. Multi scale feature fusion or hybrid feature extraction networks can be attempted.

C. SegNet-ResNet50

What worked and did not work: The training and validation losses have steadily decreased, but the validation losses have fluctuated significantly in the early stages. The training accuracy reached 82.5%, and the validation accuracy was about 80%. This performance demonstrates the adaptability of the model in complex tasks. The training and validation of IoU are relatively low, and the fluctuation of IoU validation shows that the model performs differently on different samples.

Future work: The performance of Seg_ResNet-50 can be improved by adding feature fusion layers, especially considering the fusion of multi-scale features. Combining other related tasks such as edge detection for joint learning can help improve semantic segmentation performance. Add Dropout layers or use mixed regularization methods to suppress overfitting.

For all these models, we can Use Bayesian optimization or grid search to optimize model hyperparameters, including learning rate, regularization coefficients, etc. If possible, obtain more diverse data, especially including more scene changes, to enhance the model's generalization ability. We can Consider using Ensemble Learning technology to integrate the advantages of different models to improve overall performance.

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